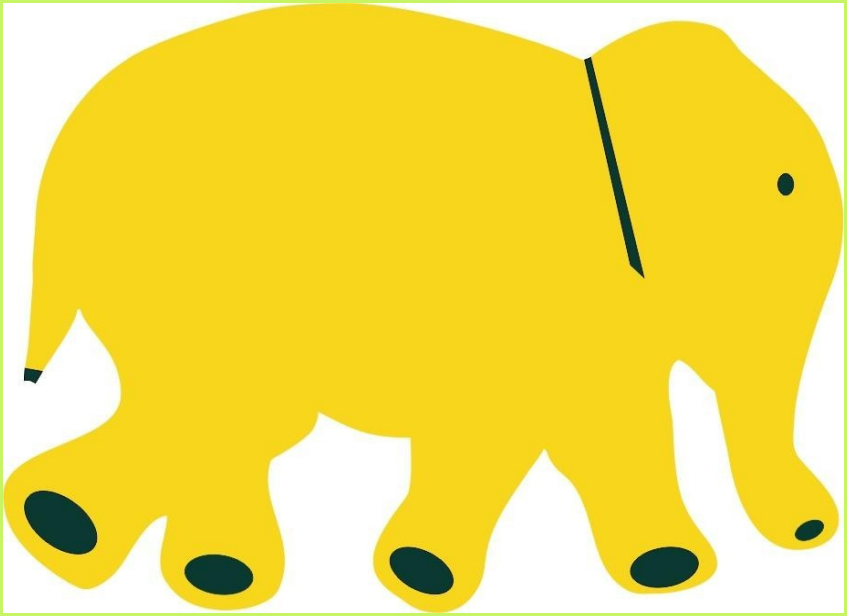


# HADOOP

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# Session Objectives

- This Session will help you to:

- Understand what is Big Data?
- List the challenges associated with Big Data
- Understand what is Hadoop?
- Understand Hadoop capabilities
- Understand Hadoop ecosystem

# What is Big Data?




- Huge Amount of Data (Terabytes or Petabytes)
- Data sets so large and complex that it becomes difficult to process.
- The challenges include capture, curation, storage, search, sharing, transfer, analysis, and visualization.




- US$ 485.651 Billion US Retailer
- Handles more than a million transactions every day, produces more 2.5 Petabytes on daily basis
- Has dedicated data centers across the world to handle the above data (Has one in Bangalore also)

- It has about a billion users as we speak
- Generates close to 500 TB of data per day
- Fires 70 thousand queries on that every day
- Inventors and one of the biggest users to hive



# The V’s of Big Data

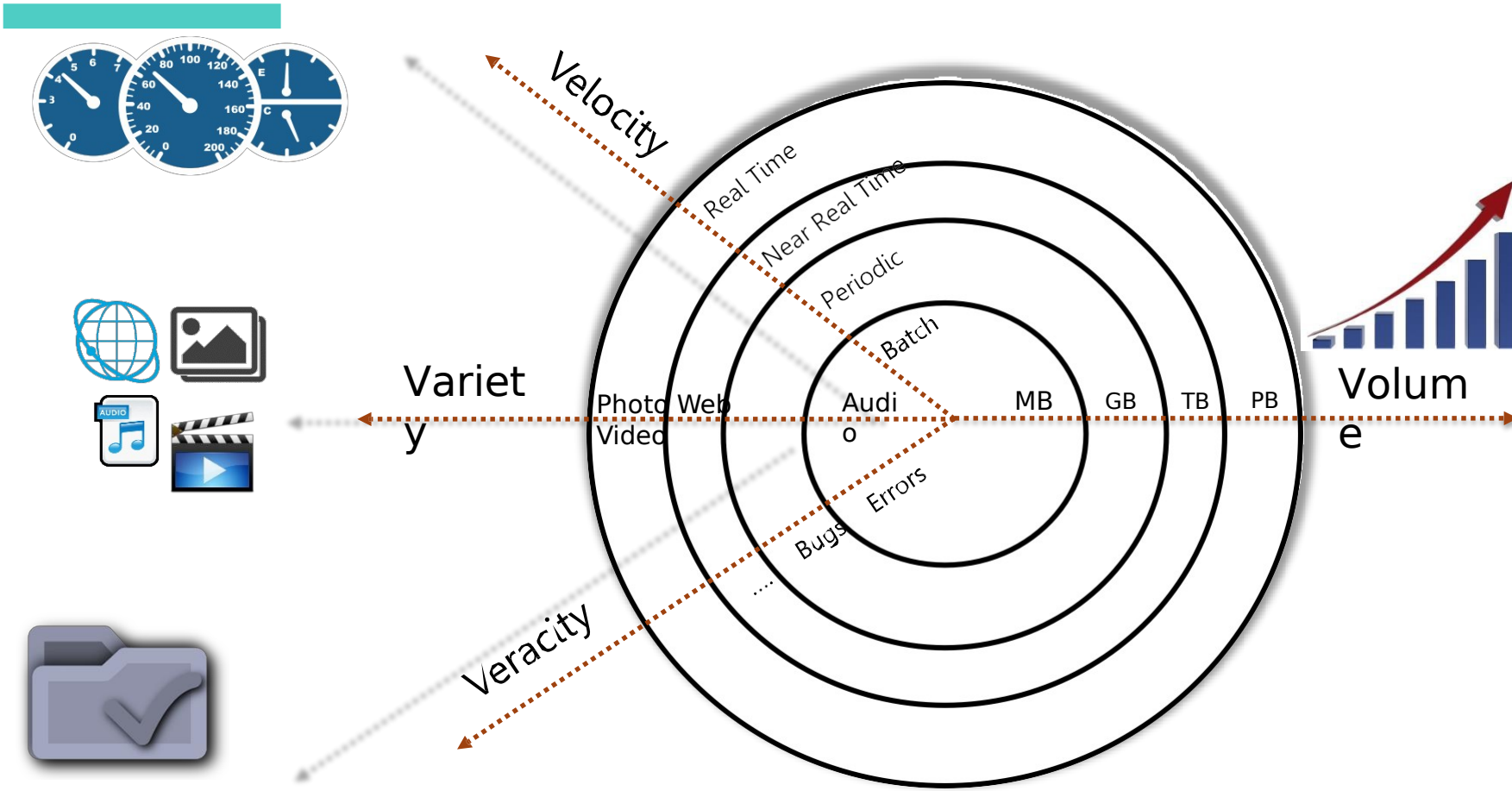
VOLUME: Refers to size of data.

VELOCITY: Speed at which data is generated, collected and analyzed.

VARIETY: Types of big data

VERACITY: Data in terms of quality and accuracy.

# Characteristics of Big Data – 4 V's



# HOW TO PROCESS BIG DATA?

# What is Hadoop?

- Apache Hadoop is a framework that allows the distributed processing of large data sets across clusters of commodity computers using a simple programming mode

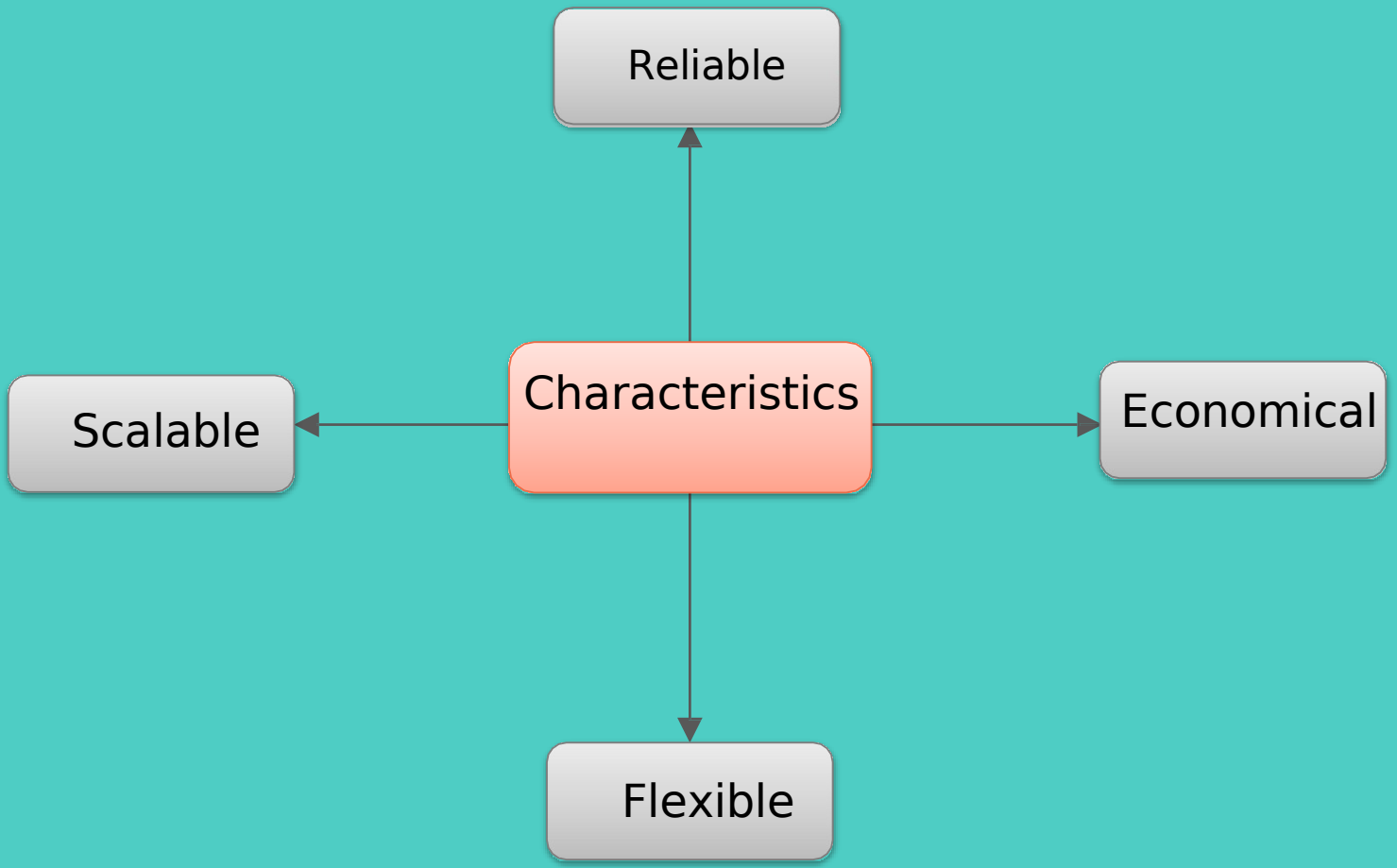



It is an Open-source Data Management with scale-out storage and "*Distributed parallel processing*".

# Hadoop Key Features

- Simple architecture
- Scalability : Designed for Massive scale
- Availability : High degree of fault tolerance; Designed to recover from Failures.
- Low Cost : Low software and hardware costs.
- Speed of Operations : Distributed file system provides fast data transfers among nodes.

# Hadoop Key Characteristics



# Hadoop Core Components




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HDFS: Data Storage frame work

Map Reduce: Data Processing Framework

# Conclusion

- Hadoop is a popular framework used for big data Storage and analytics.
- Hadoop is being applied to extremely large datasets which are otherwise complex to work using traditional database management systems.

# Thanks!

## Any queries?